System Project Requirement No. 2

Database Schema

AniMAL - A Web-Based Market Access Platform for Farmers

Direct Farm-to-Buyer Transaction Platform

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Database Management Systems (DBMS) - Systems Project Requirement

1. System and Database Name

Central database instance is named animal_marketplace and runs on MySQL 8.0+, InnoDB engine.

2. Group Members

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3. Table Summary

admin_audit_log - Before/after JSON snapshots of admin actions.
admin_order_holds - Admin intervention flags that pause an order until released.
buyer_profiles - Shopper records with delivery and pickup preferences.
farmer_profiles - Verified seller records, location, valid ID, and verification workflow.
listing_images - Photos attached to a listing; primary flag for thumbnails.
order_status_history - Immutable log of order state transitions with actor and reason.
orders - Order header holding buyer, farmer, delivery, amounts, and current status.
password_resets - One-time password-reset tokens with expiry.
produce_categories - Product taxonomy.
produce_listings - Sellable produce posts with price, stock, and lifecycle status.
quantity_units - Supported units of measure and conversion factors.
retention_holds - Legal/administrative holds preventing deletion of entities.
sessions - Active login sessions with device and IP fingerprint.
stock_adjustments - Audit trail of every stock change to a listing.
user_preferences - Per-user UI and notification settings.
user_suspensions - Account suspension records and lift audit.
users - System accounts; single sign-on identity for all roles (farmer, buyer, admin).

4. Key Mapping

Primary keys: every table has a PK - surrogate AUTO_INCREMENT ids (users, password_resets, produce_categories, quantity_units, produce_listings, listing_images, stock_adjustments, orders, order_items, order_status_history, admin_order_holds, admin_audit_log, user_suspensions, retention_holds), or natural/single-column PK (sessions.id VARCHAR(128), farmer_profiles.user_id, buyer_profiles.user_id, user_preferences.user_id).
Foreign keys reference the table and column shown in the CONSTRAINT column of the schema. ON DELETE semantics: CASCADE for child records tied to the parent identity (sessions, password_resets, listing_images, order_items, order_status_history, admin_order_holds, user_suspensions); SET NULL for audit attribution (verified_by, released_by, lifted_by); RESTRICT (default) for protected business data (buyer_id, farmer_id, listing_id, category_id, unit_id).

5. Normalization Rationale (3NF)

1NF: All column values are atomic; no repeating groups. Multi-valued data (images, status history, order line items, stock changes) are moved into separate child tables. Every table has a primary key.

2NF: No partial dependencies on the primary key. Natural-key PKs (user_id-only tables like farmer_profiles, buyer_profiles, user_preferences) depend on the whole user; composite facts stay in single-column-PK tables.
3NF: No transitive dependencies. Unit metadata (symbol, type, base_factor) lives in quantity_units and is referenced by listing.unit_id; frozen order pricing is stored in order_items.unit_price so price history does not depend on the volatile produce_listings.price_per_unit. order_items was split from orders so an order can contain several produce items without repeating order attributes in every row.

6. Database Schema (Table / Column / Data Type / Key / Null / Constraints)

Table	Column	Data Type	Key	Null	Constraints
users	id	BIGINT UNSIGNED	PK	No	AUTO_INCREMENT
users	email	VARCHAR(255)	-	No	UNIQUE
users	password_hash	VARCHAR(255)	-	No	-
users	role	ENUM('farmer','buyer','admin')	-	No	DEFAULT 'buyer'
users	status	ENUM('active','suspended','pending_verification')	-	No	DEFAULT 'pending_verification'
users	created_at	TIMESTAMP	-	No	DEFAULT CURRENT_TIMESTAMP
users	updated_at	TIMESTAMP	-	No	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP
users	last_login_at	TIMESTAMP	-	Yes	-
sessions	id	VARCHAR(128)	PK	No	-
sessions	user_id	BIGINT UNSIGNED	FK	No	References users(id) ON DELETE CASCADE
sessions	ip_address	VARCHAR(45)	-	No	-
sessions	user_agent	TEXT	-	No	-
sessions	payload	TEXT	-	No	-
sessions	last_activity	INT UNSIGNED	-	No	-
farmer_profiles	user_id	BIGINT UNSIGNED	PK	No	References users(id) ON

					DELETE CASCADE
farmer_profiles	farm_name	VARCHAR(150)	-	No	-
farmer_profiles	farm_description	TEXT	-	Yes	-
farmer_profiles	contact_person	VARCHAR(100)	-	No	-
farmer_profiles	contact_phone	VARCHAR(20)	-	Yes	-
farmer_profiles	contact_email	VARCHAR(255)	-	Yes	-
farmer_profiles	address	TEXT	-	No	-
farmer_profiles	barangay	VARCHAR(100)	-	No	-
farmer_profiles	municipality_city	VARCHAR(100)	-	No	-
farmer_profiles	province	VARCHAR(100)	-	No	-
farmer_profiles	region	VARCHAR(100)	-	No	-
farmer_profiles	latitude	DECIMAL(10,8)	-	Yes	-
farmer_profiles	longitude	DECIMAL(11,8)	-	Yes	-
farmer_profiles	valid_id_type	ENUM('national_id','drivers_license','passport','voters_id','postal_id','senior_citizen_id','pwd_id')	-	No	-
farmer_profiles	valid_id_number	VARCHAR(50)	-	No	-
farmer_profiles	valid_id_image	VARCHAR(500)	-	Yes	-
farmer_profiles	verification_status	ENUM('pending','approved','rejected')	-	No	DEFAULT 'pending'
farmer_profiles	verified_at	TIMESTAMP	-	Yes	-
farmer_profiles	verified_by	BIGINT UNSIGNED	FK	Yes	References users(id) ON DELETE SET NULL

farmer_profiles	rejection_reason	TEXT	-	Yes	-
farmer_profiles	created_at	TIMESTAMP	-	No	DEFAULT CURRENT_TIMESTAMP
farmer_profiles	updated_at	TIMESTAMP	-	No	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP
buyer_profiles	user_id	BIGINT UNSIGNED	PK	No	References users(id) ON DELETE CASCADE
buyer_profiles	full_name	VARCHAR(150)	-	No	-
buyer_profiles	contact_phone	VARCHAR(20)	-	Yes	-
buyer_profiles	delivery_address	TEXT	-	Yes	-
buyer_profiles	barangay	VARCHAR(100)	-	Yes	-
buyer_profiles	municipality_city	VARCHAR(100)	-	Yes	-
buyer_profiles	province	VARCHAR(100)	-	Yes	-
buyer_profiles	region	VARCHAR(100)	-	Yes	-
buyer_profiles	latitude	DECIMAL(10,8)	-	Yes	-
buyer_profiles	longitude	DECIMAL(11,8)	-	Yes	-
buyer_profiles	preferred_pickup	TINYINT(1)	-	No	DEFAULT 1
buyer_profiles	created_at	TIMESTAMP	-	No	DEFAULT CURRENT_TIMESTAMP
buyer_profiles	updated_at	TIMESTAMP	-	No	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP
produce_categories	id	BIGINT UNSIGNED	PK	No	AUTO_INCREMENT
produce_categories	name	VARCHAR(100)	-	No	UNIQUE
produce_categories	slug	VARCHAR(100)	-	No	UNIQUE
produce_categories	description	TEXT	-	Yes	-

produce_categories	sort_order	INT	-	No	DEFAULT 0
produce_categories	is_active	TINYINT(1)	-	No	DEFAULT 1
produce_categories	created_at	TIMESTAMP	-	No	DEFAULT CURRENT_TIMESTAMP
produce_categories	updated_at	TIMESTAMP	-	No	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP
quantity_units	id	BIGINT UNSIGNED	PK	No	AUTO_INCREMENT
quantity_units	name	VARCHAR(50)	-	No	UNIQUE
quantity_units	symbol	VARCHAR(10)	-	No	UNIQUE
quantity_units	type	ENUM('weight','count','volume')	-	No	DEFAULT 'weight'
quantity_units	base_factor	DECIMAL(10,4)	-	No	DEFAULT 1.0000
quantity_units	step	DECIMAL(10,4)	-	No	DEFAULT 0.01
quantity_units	is_active	TINYINT(1)	-	No	DEFAULT 1
produce_listings	id	BIGINT UNSIGNED	PK	No	AUTO_INCREMENT
produce_listings	farmer_id	BIGINT UNSIGNED	FK	No	References farmer_profiles(user_id) ON DELETE CASCADE
produce_listings	category_id	BIGINT UNSIGNED	FK	No	References produce_categories(id) ON DELETE RESTRICT
produce_listings	unit_id	BIGINT UNSIGNED	FK	No	References quantity_units(id) ON DELETE RESTRICT
produce_listings	name	VARCHAR(150)	-	No	-
produce_listings	description	TEXT	-	Yes	-

produce_listings	price_per_unit	DECIMAL(12,2)	-	No	-
produce_listings	available_quantity	DECIMAL(12,4)	-	No	DEFAULT 0.0000
produce_listings	minimum_order_qty	DECIMAL(12,4)	-	No	DEFAULT 0.0100
produce_listings	maximum_order_qty	DECIMAL(12,4)	-	Yes	-
produce_listings	harvest_date	DATE	-	Yes	-
produce_listings	expiry_date	DATE	-	Yes	-
produce_listings	is_organic	TINYINT(1)	-	No	DEFAULT 0
produce_listings	status	ENUM('draft','active','paused','sold_out','archived')	-	No	DEFAULT 'draft'
produce_listings	rejection_reason	TEXT	-	Yes	-
produce_listings	created_at	TIMESTAMP	-	No	DEFAULT CURRENT_TIMESTAMP
produce_listings	updated_at	TIMESTAMP	-	No	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP
produce_listings	published_at	TIMESTAMP	-	Yes	-
listing_images	id	BIGINT UNSIGNED	PK	No	AUTO_INCREMENT
listing_images	listing_id	BIGINT UNSIGNED	FK	No	References produce_listings(id) ON DELETE CASCADE
listing_images	image_path	VARCHAR(500)	-	No	-
listing_images	is_primary	TINYINT(1)	-	No	DEFAULT 0
listing_images	sort_order	INT	-	No	DEFAULT 0
listing_images	created_at	TIMESTAMP	-	No	DEFAULT CURRENT_TIMESTAMP

stock_adjustments	id	BIGINT UNSIGNED	PK	No	AUTO_INCREMENT
stock_adjustments	listing_id	BIGINT UNSIGNED	FK	No	References produce_listings(id) ON DELETE CASCADE
stock_adjustments	adjustment_type	ENUM('initial','restock','sale_deduction','return','spoilage','admin_correction')	-	No	-
stock_adjustments	quantity_change	DECIMAL(12,4)	-	No	-
stock_adjustments	quantity_before	DECIMAL(12,4)	-	No	-
stock_adjustments	quantity_after	DECIMAL(12,4)	-	No	-
stock_adjustments	reference_type	ENUM('order','manual','admin')	-	Yes	-
stock_adjustments	reference_id	BIGINT UNSIGNED	-	Yes	-
stock_adjustments	notes	TEXT	-	Yes	-
stock_adjustments	created_by	BIGINT UNSIGNED	FK	No	References users(id) ON DELETE RESTRICT
stock_adjustments	created_at	TIMESTAMP	-	No	DEFAULT CURRENT_TIMESTAMP
orders	id	BIGINT UNSIGNED	PK	No	AUTO_INCREMENT
orders	order_number	VARCHAR(30)	-	No	UNIQUE
orders	buyer_id	BIGINT UNSIGNED	FK	No	References buyer_profiles(user_id) ON DELETE RESTRICT
orders	farmer_id	BIGINT UNSIGNED	FK	No	References farmer_profiles(user_id) ON DELETE RESTRICT

orders	listing_id	BIGINT UNSIGNED	FK	No	References produce_listings(id) ON DELETE RESTRICT
orders	quantity	DECIMAL(12,4)	-	No	-
orders	unit_price	DECIMAL(12,2)	-	No	-
orders	delivery_fee	DECIMAL(12,2)	-	No	DEFAULT 0.00
orders	delivery_method	ENUM('pickup','farmer_delivery')	-	No	DEFAULT 'pickup'
orders	delivery_address	TEXT	-	Yes	-
orders	preferred_pickup_date	DATE	-	Yes	-
orders	preferred_pickup_time	TIME	-	Yes	-
orders	buyer_note	TEXT	-	Yes	-
orders	farmer_note	TEXT	-	Yes	-
orders	admin_note	TEXT	-	Yes	-
orders	status	ENUM('pending','delivery_fee_review','confirmed','preparing','ready_for_pickup','out_for_delivery','completed','rejected','cancelled')	-	No	DEFAULT 'pending'
orders	rejection_reason	TEXT	-	Yes	-
orders	cancellation_reason	TEXT	-	Yes	-
orders	cancelled_by_role	ENUM('buyer','farmer','admin')	-	Yes	-
orders	confirmed_at	TIMESTAMP	-	Yes	-
orders	preparing_at	TIMESTAMP	-	Yes	-
orders	ready_at	TIMESTAMP	-	Yes	-
orders	out_for_delivery_at	TIMESTAMP	-	Yes	-
orders	completed_at	TIMESTAMP	-	Yes	-
orders	rejected_at	TIMESTAMP	-	Yes	-
orders	cancelled_at	TIMESTAMP	-	Yes	-

orders	created_at	TIMESTAMP	-	No	DEFAULT CURRENT_TIMESTAMP
orders	updated_at	TIMESTAMP	-	No	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP
order_status_history	id	BIGINT UNSIGNED	PK	No	AUTO_INCREMENT
order_status_history	order_id	BIGINT UNSIGNED	FK	No	References orders(id) ON DELETE CASCADE
order_status_history	from_status	ENUM('pending','delivery_fee_review','confirmed','preparing','ready_for_pickup','out_for_delivery','completed','rejected','cancelled')	-	Yes	-
order_status_history	to_status	ENUM('pending','delivery_fee_review','confirmed','preparing','ready_for_pickup','out_for_delivery','completed','rejected','cancelled')	-	No	-
order_status_history	changed_by	BIGINT UNSIGNED	FK	No	References users(id) ON DELETE RESTRICT
order_status_history	changed_by_role	ENUM('buyer','farmer','admin')	-	No	-
order_status_history	reason	TEXT	-	Yes	-

order_status_history	created_at	TIMESTAMP	-	No	DEFAULT CURRENT_TIMESTAMP
admin_order_holds	id	BIGINT UNSIGNED	PK	No	AUTO_INCREMENT
admin_order_holds	order_id	BIGINT UNSIGNED	FK	No	References orders(id) ON DELETE CASCADE
admin_order_holds	admin_id	BIGINT UNSIGNED	FK	No	References users(id) ON DELETE RESTRICT
admin_order_holds	reason	TEXT	-	No	-
admin_order_holds	is_active	TINYINT(1)	-	No	DEFAULT 1
admin_order_holds	created_at	TIMESTAMP	-	No	DEFAULT CURRENT_TIMESTAMP
admin_order_holds	released_at	TIMESTAMP	-	Yes	-
admin_order_holds	released_by	BIGINT UNSIGNED	FK	Yes	References users(id) ON DELETE SET NULL
user_suspensions	id	BIGINT UNSIGNED	PK	No	AUTO_INCREMENT
user_suspensions	user_id	BIGINT UNSIGNED	FK	No	References users(id) ON DELETE CASCADE
user_suspensions	admin_id	BIGINT UNSIGNED	FK	No	References users(id) ON DELETE RESTRICT
user_suspensions	reason	TEXT	-	No	-
user_suspensions	suspended_at	TIMESTAMP	-	No	DEFAULT CURRENT_TIMESTAMP
user_suspensions	lifted_at	TIMESTAMP	-	Yes	-
user_suspensions	lifted_by	BIGINT UNSIGNED	FK	Yes	References users(id) ON

					DELETE SET NULL
retention_holds	id	BIGINT UNSIGNED	PK	No	AUTO_INCREMENT
retention_holds	entity_type	ENUM('user','order','listing','verification')	-	No	-
retention_holds	entity_id	BIGINT UNSIGNED	-	No	-
retention_holds	admin_id	BIGINT UNSIGNED	FK	No	References users(id) ON DELETE RESTRICT
retention_holds	reason	TEXT	-	No	-
retention_holds	expires_at	TIMESTAMP	-	Yes	-
retention_holds	created_at	TIMESTAMP	-	No	DEFAULT CURRENT_TIMESTAMP
retention_holds	released_at	TIMESTAMP	-	Yes	-
retention_holds	released_by	BIGINT UNSIGNED	FK	Yes	References users(id) ON DELETE SET NULL
admin_audit_log	id	BIGINT UNSIGNED	PK	No	AUTO_INCREMENT
admin_audit_log	admin_id	BIGINT UNSIGNED	FK	No	References users(id) ON DELETE RESTRICT
admin_audit_log	action	VARCHAR(100)	-	No	-
admin_audit_log	entity_type	VARCHAR(50)	-	No	-
admin_audit_log	entity_id	BIGINT UNSIGNED	-	Yes	-
admin_audit_log	old_values	JSON	-	Yes	-
admin_audit_log	new_values	JSON	-	Yes	-
admin_audit_log	ip_address	VARCHAR(45)	-	Yes	-

admin_audit_log	user_agent	TEXT	-	Yes	-
admin_audit_log	created_at	TIMESTAMP	-	No	DEFAULT CURRENT_TIMESTAMP